Day : 09

* UDP – User Datagram Protocol

- UDP is a connectionless, unreliable transport layer protocol (Layer 4).

* TCP vs UDP

Feature	TCP	UDP
Connection	3-way handshake	No handshake
Reliability	Guaranteed delivery	No guarantee
Speed	Slower	Faster
Order	In Order	No Order
Use Case	web, email, file transfer	Video, DNS, VoIP, gaming

* UDP Header (Only 8 bytes)

Field	Size
Source Port	16 bits
Destination Port	16 bits
Length	16 bits
Checksum	16 bits

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Note: No sequence numbers, no ACK, no flow control.

* UDP in DFIR

- DNS uses UDP (port 53) — most DNS queries are UDP.
- DHCP uses UDP (port 67/68).
- Attackers use UDP for UDP flood attacks (DoS)
- Malware can use UDP to avoid TCP connection tracking.
- Wireshark filter: udp

* ICMP — Internet Control Message Protocol

- ICMP is a network layer protocol (Layer 3) used for diagnostics and error reporting. It is NOT used to transfer data — only control messages.

* Key ICMP Types:

Type	Name	Purpose
0	Echo Reply	Response to ping
3	Destination Unreachable	Host/port not

	reachable
8 Echo Request	The ping itself
11 Time Exceeded	TTL expired (traceroute)

* How ping works:

Your PC

↑ ↓

(Type 0) ICMP Echo Request (Type 8)

↑ ↓

google.com

* ICMP in DFIR

- Ping Sweep — attacker pings many IPs to find live hosts.
- ICMP Tunneling — data hidden inside ICMP packets for exfiltration.
- TTL Exceeded — used by traceroute, shows network path.
- High ICMP volume = possible reconnaissance.
- Wireshark filter: icmp

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* ARP ← Address Resolution Protocol

- ARP (Address Resolution Protocol) operates at Layer 2 (Data Link). It translates IP addresses → MAC addresses within a local network.

* • Why is this needed? *

- IP = logical address (used for routing)
- MAC = physical address (used for actual delivery on local network)
- Before sending data, your PC must know the MAC address of the destination.

* ARP Request vs Reply:

- PC - A broadcasts: "who has 192.168.1.1? Tell 192.168.1. 50".
- Router replies: "192.168.1.1 is at AA:BB:CC:DD:EE:FF".

* Message	Direction	Purpose
ARP Request	Broadcast (to everyone)	who has this IP?
ARP Reply	Unicast (to requester)	I have that IP, here's my MAC

Note: ARP Cache — your PC stores IP → MAC mapping temporarily to avoid repeated ARP requests.

* ARP in DFIR

- ARP Poisoning / Spoofing — attacker sends fake ARP replies to poison ARP cache.

- This causes traffic meant for router to go to attacker insted — Man in the Middle attack.

- Signs: same MAC claiming multiple IPs, high ARP rate.

- Gratuitous ARP — device announces its own IP → MAC without being asked (suspicious!)

- Wireshark filter: arp

* Diagrams:

- UDP vs ICMP vs ARP

UDP	ICMP
Layer 4 — no handshake	Layer 3 — diagnostics

ARP
Layer 2 — IP → MAC

- UDP — Connectionless

No handshake
Data Sent Directly

8-byte header only
Src / Dst port, len, checksum

Common ports
DNS : 53 , DHCP: 67/68

- ICMP — ping protocol

Your PC —Type 8→ Server
Your PC ←Type 0— Server

ICMP types
Type 8 = Echo Request
Type 0 = Echo Reply
Type 11 = TTL Exceeded
Type 3 = Destination Unreachable

- ARP — IP to MAC

ARP Request
Broadcast — who has X.X.X.X?

↓

ARP Reply
Unicast — MAC = XX:XX:XX

ARP Cache

* DFIR — Attack Patterns

UDP Flood
DoS via UDP packets

ICMP tunneling
Data hidden in ping packets

ARP Poisoning
Fake ARP → MITM attack

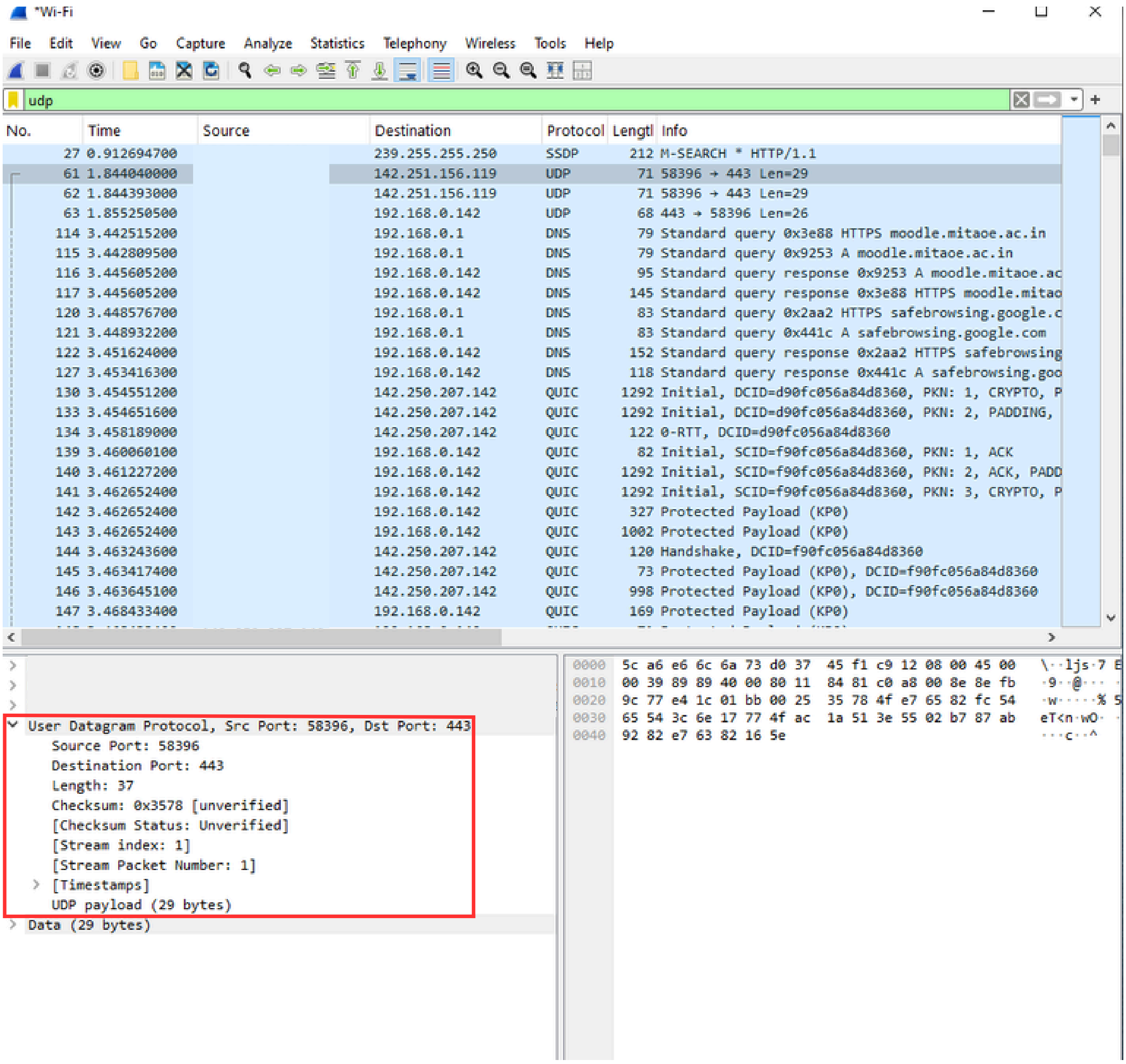
• Wireshark Filters

udp

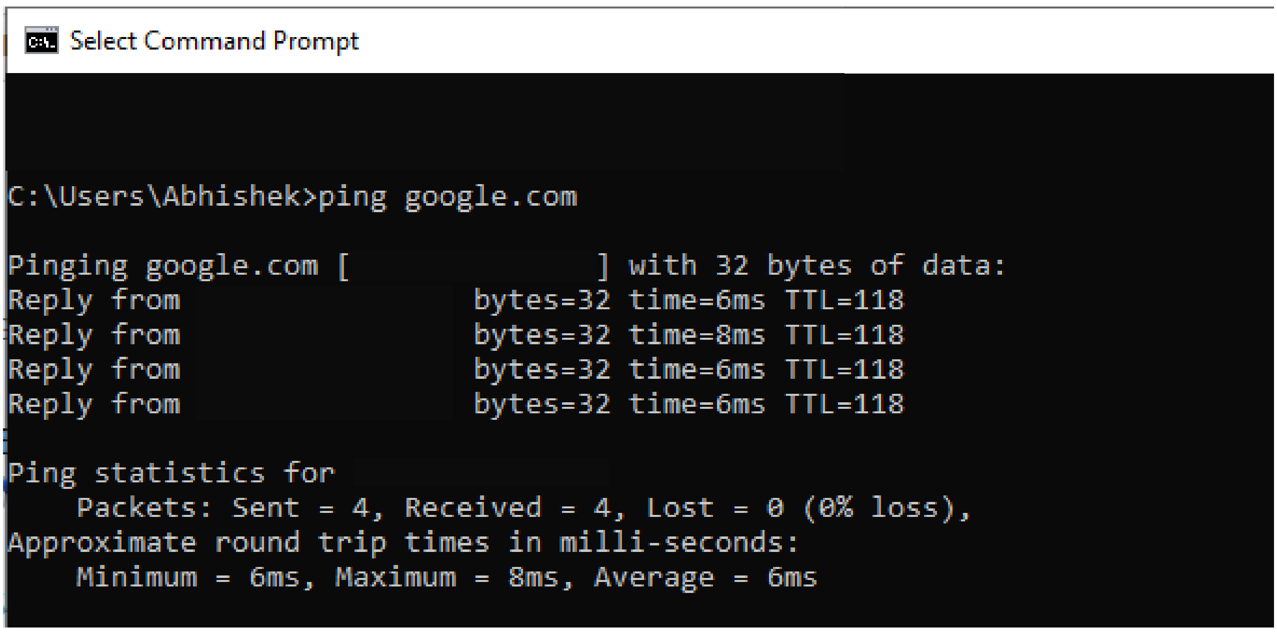
icmp

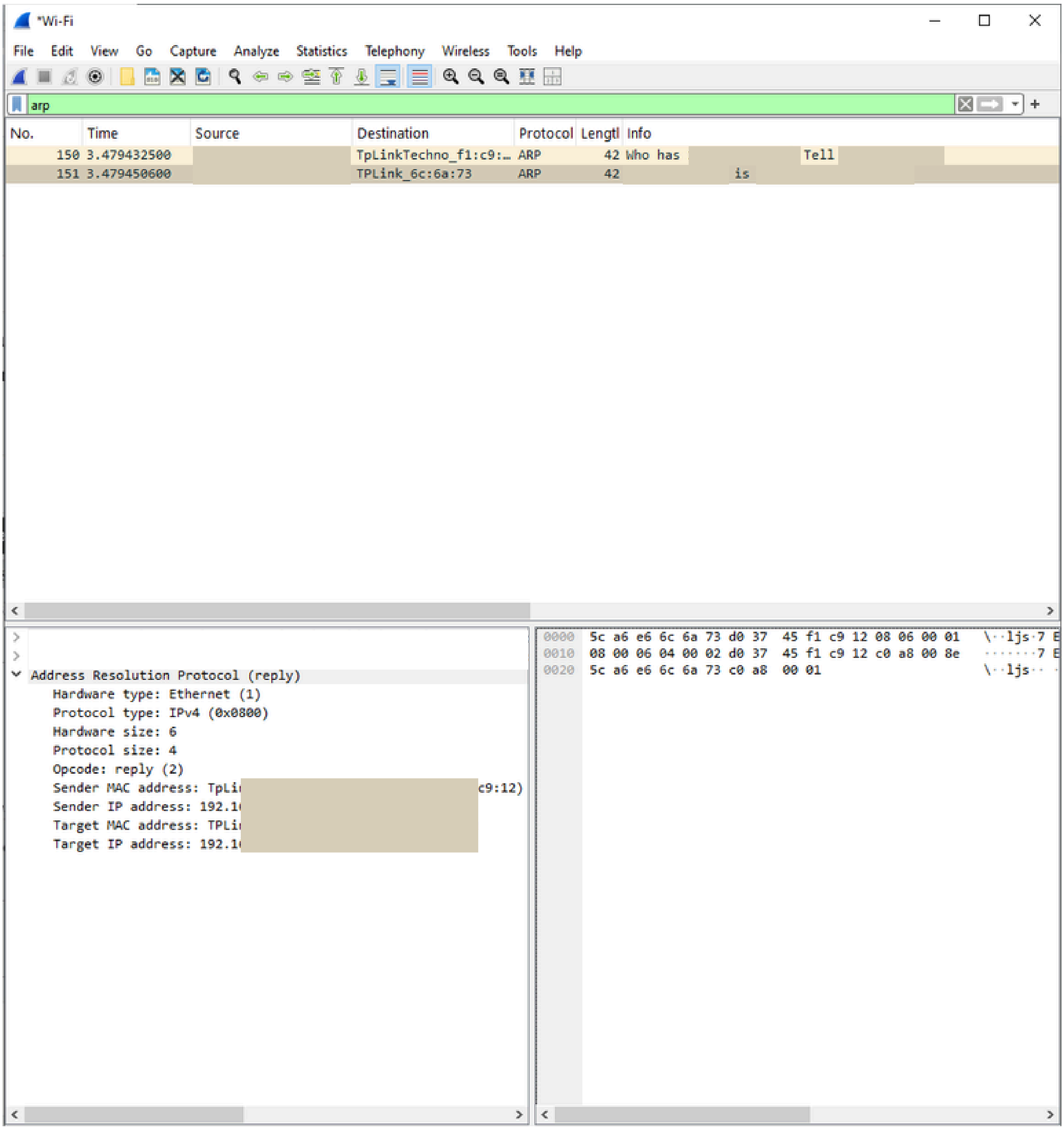
arp

* TCP = connection-oriented
* UDP = connectionless
* ICMP = diagnostic
* ARP = local resolution


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
No. Time Source Destination Protocol Length Info
27 0.912694700 239.255.255.250 SSDP 212 M-SEARCH * HTTP/1.1
61 1.844040000 142.251.156.119 UDP 71 58396 → 443 Len=29
62 1.844393000 142.251.156.119 UDP 71 58396 → 443 Len=29
63 1.855250500 192.168.0.142 UDP 68 443 → 58396 Len=26
114 3.442515200 192.168.0.1 DNS 79 Standard query 0x3e88 HTTPS moodle.mitaoe.ac.in
115 3.442809500 192.168.0.1 DNS 79 Standard query 0x9253 A moodle.mitaoe.ac.in
116 3.445605200 192.168.0.142 DNS 95 Standard query response 0x9253 A moodle.mitaoe.ac
117 3.445605200 192.168.0.142 DNS 145 Standard query response 0x3e88 HTTPS moodle.mitao
120 3.448576700 192.168.0.1 DNS 83 Standard query 0x2aa2 HTTPS safebrowsing.google.c
121 3.448932200 192.168.0.1 DNS 83 Standard query 0x441c A safebrowsing.google.com
122 3.451624000 192.168.0.142 DNS 152 Standard query response 0x2aa2 HTTPS safebrowsing
127 3.453416300 192.168.0.142 DNS 118 Standard query response 0x441c A safebrowsing.goo
130 3.454551200 142.250.207.142 QUIC 1292 Initial, DCID=d90fc056a84d8360, PKN: 1, CRYPTO, P
133 3.454651600 142.250.207.142 QUIC 1292 Initial, DCID=d90fc056a84d8360, PKN: 2, PADDING,
134 3.458189000 142.250.207.142 QUIC 122 0-RTT, DCID=d90fc056a84d8360
139 3.460060100 192.168.0.142 QUIC 82 Initial, SCID=f90fc056a84d8360, PKN: 1, ACK
140 3.461227200 192.168.0.142 QUIC 1292 Initial, SCID=f90fc056a84d8360, PKN: 2, ACK, PADD
141 3.462652400 192.168.0.142 QUIC 1292 Initial, SCID=f90fc056a84d8360, PKN: 3, CRYPTO, P
142 3.462652400 192.168.0.142 QUIC 327 Protected Payload (KP0)
143 3.462652400 192.168.0.142 QUIC 1002 Protected Payload (KP0)
144 3.463243600 142.250.207.142 QUIC 120 Handshake, DCID=f90fc056a84d8360
145 3.463417400 142.250.207.142 QUIC 73 Protected Payload (KP0), DCID=f90fc056a84d8360
146 3.463645100 142.250.207.142 QUIC 998 Protected Payload (KP0), DCID=f90fc056a84d8360
147 3.468433400 192.168.0.142 QUIC 169 Protected Payload (KP0)
User Datagram Protocol, Src Port: 58396, Dst Port: 443
Source Port: 58396
Destination Port: 443
Length: 37
Checksum: 0x3578 [unverified]
[Checksum Status: Unverified]
[Stream index: 1]
[Stream Packet Number: 1]
[Timestamps]
UDP payload (29 bytes)
Data (29 bytes)
0000 5c a6 e6 6c 6a 73 d0 37 45 f1 c9 12 08 00 45 00
0010 00 39 89 89 40 00 80 11 84 81 c0 a8 00 8e 8e fb
0020 9c 77 e4 1c 01 bb 00 25 35 78 4f e7 65 82 fc 54
0030 65 54 3c 6e 17 77 4f ac 1a 51 3e 55 02 b7 87 ab
0040 92 82 e7 63 82 16 5e

ICMP


Select Command Prompt
C:\Users\Abhishek>ping google.com
Pinging google.com [ ] with 32 bytes of data:
Reply from bytes=32 time=6ms TTL=118
Reply from bytes=32 time=8ms TTL=118
Reply from bytes=32 time=6ms TTL=118
Reply from bytes=32 time=6ms TTL=118
Ping statistics for
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 6ms, Maximum = 8ms, Average = 6ms


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
arp
No. Time Source Destination Protocol Lengtl Info
150 3.479432500 TpLinkTechno_f1:c9:… ARP 42 Who has Tell
151 3.479450600 TPLink_6c:6a:73 ARP 42 is
Address Resolution Protocol (reply)
Hardware type: Ethernet (1)
Protocol type: IPv4 (0x0800)
Hardware size: 6
Protocol size: 4
Opcode: reply (2)
Sender MAC address: TpLi c9:12)
Sender IP address: 192.1
Target MAC address: TPLi
Target IP address: 192.1
0000 5c a6 e6 6c 6a 73 d0 37 45 f1 c9 12 08 06 00 01
0010 08 00 06 04 00 02 d0 37 45 f1 c9 12 c0 a8 00 8e
0020 5c a6 e6 6c 6a 73 c0 a8 00 01